PONDICHERRY UNIVERSITY
M.Sc. QUANTITATIVE FINANCE
(CHOICE BASED CREDIT SYSTEM)
Academic Year 2025-26 onwards

Non - Credit Bridge Courses (covering various concepts in bullet points)			Nature of the Course
Pre Semester	MSQF 401	Basics of Business and Accounting	Compulsory Bridge Courses for 2 weeks
	MSQF 402	Basics of Computer Programming	
	MSQF 403	Basics of Economics	
	MSQF 404	Basics of Research Methods	

Semester	Course Code	Title of the Course	Nature of the Course	No. of Credits
I	MSQF 411	Accounting and Financial Analysis	Hard Core	4
	MSQF 412	Financial Management	Hard Core	4
	MSQF 413	Managerial Economics	Hard Core	4
	MSQF 414	Statistical Methods	Hard Core	4
	MSQF 415	Lab I:Financial Statement Analysis (Using Spreadsheet)	Hard Core	2
	MSQF 416	Lab II: Data Analysis Using Python	Hard Core	2
		Any one Soft Core paper among the list of soft core courses / Relevant courses can be opted from SWAYAM/NPTEL/CFA	Soft Core	3
II	MSQF 421	Investment Management	Hard Core	4
	MSQF 422	Optimization Methods	Hard Core	4
	MSQF 423	Econometrics	Hard Core	4
	MSQF 424	Financial Institutions and Markets	Hard Core	4
	MSQF 425	Lab III: Data Analytics using R	Hard Core	2
	MSQF 426	Lab IV: CAPSTONE Simulation Lab.	Hard Core	2
		Any one Soft Core paper among the list of soft core courses / Relevant courses can be opted from SWAYAM/NPTEL/CFA	Soft Core	3
III	MSQF 531	Applied Time Series Analysis and Forecasting	Hard Core	4
	MSQF 532	Financial Mathematics	Hard Core	4
	MSQF 533	Financial Risk Management	Hard Core	4
	MSQF 534	Corporate Internship	Hard Core	6
	MSQF 535	Global Financial Management	Hard Core	4
	MSQF 536	Machine Learning Techniques	Hard Core	4
		Any two Soft Core paper among the list of soft core courses/Relevant courses can be opted from SWAYAM/NPTEL/CFA	Soft Core Soft Core	3 3
IV	MSQF 541	Project Work cum Internship	Hard Core	12
		List of Soft Core Courses		
	MSQF 417	Behavioural Finance	Soft Core	3
	MSQF 418	Quantum Finance	Soft Core	3
	MSQF 419	Micro Finance and Entrepreneurship Management	Soft Core	3
	MSQF 427	Research Methods for Finance	Soft Core	3
	MSQF 428	Investment and Commercial Banking	Soft Core	3
	MSQF 429	Infrastructure Finance	Soft Core	3
	MSQF 537	Block Chain Technology for Finance	Soft Core	3
	MSQF 538	Stochastic Calculus for Finance	Soft Core	3
	MSQF 539	Fintech and Sustainable Finance	Soft Core	3
	MSQF 540	Big Data Analytics and Cloud Computing	Soft Core	3
	MSQF 542	Financial Engineering and Derivatives	Soft Core	3

MSQF 531: APPLIED TIME SERIES ANALYSIS AND FORECASTING CREDITS:4

Course Objectives:

- *Providing a clear explanation of the fundamental theory of time series analysis and forecasting*
- *The book features treatments of forecast improvement with regression and autoregression combination models and model and forecast evaluation, along with a sample size analysis for common time series models to attain adequate statistical power*

Course Outcomes (COs):

CO1: Understand and interpret the fundamental concepts and components of time series data
CO2: Apply smoothing techniques for real world data using R and Excel
CO3: Distinguish between stationary and non-stationary time series,
CO4: Application and modeling time-varying volatility using financial data
CO5: Evaluate and compare various forecasting methods using real time data

Unit I: Introduction to Time Series

Definition and examples of Time Series Models-Graphical Representation of Time Series Data -Approaches of Time Series –Additive and multiplicative approach-Components and various decompositions of Time Series Models-Numerical description of Time Series - Data transformations - Methods of estimation –Trend, Seasonal and exponential.

Unit II: Smoothing Techniques and univariate Time Series Models

Smoothing Techniques: Moving Averages: Simple, centered, double and weighted moving averages; single and double exponential smoothing – Holt's and winter's methods - Exponential smoothing techniques for series with trend and seasonality- First and Second order AR and MA Models – ARMA/ARIMA Models -Mixed ARMA /ARIMA models their statistical Properties – box Jenkins methodology ACF and PAF functions-Finite order AR(p) and MA(q) models .

Unit III: Multivariate and Multiple Equation Models

Trend stationary -Stationary Time Series Models –General Unit Root Tests: Dickey Fuller Test, Augmented Dickey Fuller Test - Johansen Test for cointegration - Engle Granger causality - error correction model -Vector Autoregressive (VAR) model - Vector Error Correction Model (VECM), estimation of Lag models.

Unit IV: Modeling Volatility

Impulse response function, variance decomposition - Definition and representation of ARCH and GARCH Models-their use in financial time series data- Volatility forecasting using GARCH (1,1) Model- Diagnostic checking of model – analysis of residuals.

Unit V: Applications and Forecast Evaluation

Introduction to business forecasting –scope-types of forecasting- Forecasting cycle-different forecasting techniques-Exploring data patterns and choosing forecasting technique- Managing forecasting process- measuring forecasting error -Forecasting error comparison -Forecast accuracy measures (MAE, MSE, RMSE, MAPE)-Model validation and backtesting -Scenario analysis and stress testing in financial forecasting-Case studies: stock prices, interest rates, macroeconomic forecasting-Use of software tools (R, Python) for applied time series modeling

Books for Reference

1. Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.).
2. Hamilton, J. D. (2023). *Time series analysis* (2nd ed.). Princeton University Press.
3. George E. P. Box, G. M. Jenkins and G. C. Reinsel, (1994) :Time Series analysis Forecasting and Control, Prentice Hall International, 3/e . (Unit II and IV)
4. Makridakis. S. and Wheelwright. S. C. (1997): Forecasting: Methods and Applications, 3/e John Wiley & Sons.
5. Mills, T., (1997): The Econometric Modeling of Financial Time Series.2/e Cambridge University Press.
6. Shumway & Stoffer, D. S. (2023). *Time series analysis and its applications: With R examples* (4th ed.). Springer.
7. Tsay, R. S. (2024). *Analysis of financial time series* (4th ed.). Wiley.Montgomery D.C., C.L.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XX	XX	XX	XX
CO2	XXX	XXX	XX	XX	XX
CO3	XXX	XX	XX	XX	XX
CO4	XXX	XXX	XXX	XX	XX
CO5	XXX	XXX	XXX	XX	XXX

MSQF 532: FINANCIAL MATHEMATICS **CREDITS: 4**

Course Objectives:

- *To equip students with a solid understanding of fundamental mathematical concepts and techniques used in financial analysis, valuation of financial instruments, and portfolio management.*
- *To develop the ability to apply quantitative methods for solving practical problems in finance, including risk assessment, derivatives pricing, and financial decision-making.*

Course Outcomes (COs):

CO1: Apply financial mathematics to evaluate investments and corporate finance decisions
CO2: Use quantitative tools for bond, equity, and portfolio valuation
CO3: Analyze risk and return to support financial strategy
CO4: Interpret and construct financial statements for cash flow management
CO5: Apply mathematical models to real-world financial problems

Unit I: Introduction to Financial Mathematics

Role and importance of mathematics in finance -Time value of money concepts-Simple interest and compound interest-Nominal and effective rates-Discounting and present value-Basic annuities: ordinary annuities, annuities due- Growth and decay curves

Unit II: Valuation of Financial Instruments

Present value and future value of annuities-Perpetuities and growing annuities Loan amortization and sinking funds-Bond valuation: yield to maturity (YTM), yield to call (YTC)-Equity valuation: dividend discount model (DDM), price-earnings model

Unit III: Financial Ratios and Cash Flow Analysis

Key financial ratios: liquidity, profitability, leverage, efficiency-Application of ratios in decision making-Cash flow statements: operating, investing, and financing cash flows-Free cash flow analysis-Break-even and sensitivity analysis

Unit IV: Risk, Return, and Portfolio Mathematics

Measurement of return: arithmetic vs geometric mean-Standard deviation and variance of returns-Coefficient of variation-Covariance and correlation-Portfolio return and risk-Diversification and optimal portfolio construction-Capital Asset Pricing Model (CAPM)

Unit V: Advanced Topics and Applications

Linear programming in financial decision-making -Introduction to financial derivatives (options, forwards, futures)-Option valuation basics: Binomial and Black-Scholes models-Duration and convexity of bonds-Probability distributions in finance: normal, log-normal-Simulation and scenario analysis in financial planning

Books for Reference:

1. Bodie, Z., Kane, A., & Marcus, A. J. (2022). Investments (12th ed.). McGraw-Hill Education.
2. Tuckman, B., & Serrat, A. (2023). Fixed Income Securities: Tools for Today's Markets (4th ed.). Wiley.
3. Fabozzi, F. J. (2023). Financial Mathematics: A Comprehensive Treatment (2nd ed.). Wiley.
4. Hull, J. C. (2024). Options, Futures, and Other Derivatives (11th ed.). Pearson.
5. Ross, S. A., Westerfield, R. W., Jaffe, J., & Jordan, B. D. (2022). Corporate Finance (13th ed.). McGraw-Hill Education.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XXX	XXX	XX	XX
CO2	XXX	XXX	XXX	XX	XX
CO3	XXX	XX	XXX	XXX	XX
CO4	XX	XX	XX	XXX	XX
CO5	XXX	XXX	XXX	XX	XXX

MSQF 533: FINANCIAL RISK MANAGEMENT CREDITS: 4

Course Objectives:

- *This paper focuses the basic concept of risk management and expose various types of risk faced*
- *It helps to take positions for investing and trading in options and future market*

Course Outcomes (COs):

CO1: Analyse how futures and forward markets operate and be able to calculate its theoretical prices and values
CO2: Analyse the sources of financial risk and the importance of implementing effective financial risk management
CO3: Evaluate hedging strategies using forwards, futures, options and swaps
CO4: to identify financial risks in currencies, interest rates, commodities and shares
CO5: to evaluate the outcomes of strategies to manage Risk with the help of case Studies

Unit I: Introduction to Risk Management

Introduction to risk management- Sources of risk- risks of commodity investment - risks of equity investment - risks of fixed income investment- risks of currency investment-market risk measurement

Unit II: VAR Methods and Hedging

An overview of VAR- definition, downside VAR - VAR methods -VAR local and full valuation, delta normal methods, historical Simulation, Monte Carlo simulation, VAR applications - stress testing, Hedging: liner risk, optimal hedging, hedge ratio as regression coefficient, non-linear risk hedging.

Unit III: Credit Risk Management and

Introduction to credit risk - measuring credit risk- credit exposure - types of credit Derivatives- credit default swap - pricing and hedging credit derivatives- credit risk models- Basel accord- the Basel market risk charges, Settlement risk.

Unit: IV Management of Financial Risk

Measuring & Managing Interest Rate Risks – Discounted Cash Flow, Time Value of Money, Yield to Maturity, Fixed Term Annuities & Present Value of Basis Point - Measuring & Managing Market Risk – Risk/Return Trade-off, Systematic Risk, Idiosyncratic Risk, Modern Portfolio Theorem & Risk Adjusted Return on Capital (RAROC) - Measuring & Managing Liquidity Risk: Forecasting cash requirements from Cash Flow Analysis - Working Capital Management –

Unit V: Operation & Integrated Risk Management

Measuring & Managing Operational Risk - Brainstorming, SWOT, Root Cause Analysis & Assumption Analysis - Measuring & Managing Credit & Counterparty Risk – Syndication of Credit, Netting of Payments (Cross Collateralization), Third Party Guarantees (Banks) & Credit Derivatives - Measuring Systemic & Sovereign Risk – Assessing Debt/GDP, Debt/Revenue, Debt Servicing, BoP Position and FOREX Reserves & Demographic Profile.

Books for Reference :

1. Crouhy, M., Galai, D., & Mark, R. (2023). *The essentials of risk management* (3rd ed.). McGraw-Hill.
2. Hopkin, P., & Thompson, C. (2022). *Fundamentals of risk management: Understanding, evaluating and implementing effective enterprise risk management* (6th ed.). Kogan Page Ltd.
3. Lee, H. (2021). *Risk management: Fundamentals, theory, and practice in Asia*. Springer Singapore. https://doi.org/10.1007/978-981-16-3467-3
4. Hunziker, S. (2021). *Enterprise risk management: Modern approaches to balancing risk and reward*. Springer Gabler. ISBN: 978-3658335229.
5. Indian Institute of Banking & Finance (IIBF). (2023). *Risk management*. Macmillan.
6. Jorion, P. (2003): Financial Risk Management Handbook, Wiley.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XXX	XX	XX	X
CO2	XXX	XX	XX	XXX	XX
CO3	XXX	XXX	XXX	XX	XX
CO4	XX	XX	XX	XXX	X
CO5	XX	XX	XXX	XXX	XX

MSQF 534: CORPORATE INTERNSHIP AND VIVA CREDITS: 6

Course Objectives:

- *Internship consists of an exchange of services for experience between the student and an organization*.
- *The purpose of the student internship is to provide an opportunity to seek, identify and further develop an appropriate level of professionalism .*
- *To expand network of professional relationships and contacts.*

COURSE OUTCOMEs (COs):

CO1: Demonstrate the ability to integrate academic knowledge with practical applications by undertaking real-world financial tasks in banks, financial institutions, stock markets, investment banks, insurance, merchant banking, or broking firms.
CO2: Apply quantitative, statistical, and analytical techniques learned during the coursework to analyze financial products, services, and markets.
CO3: Develop industry-relevant skills including financial modeling, data analysis, report writing, and business communication through structured internship activities.
CO4: Exhibit professionalism, teamwork, and ethical responsibility by working under the supervision of company and academic guides and adhering to industry protocols.
CO5: Prepare and present a comprehensive internship report and presentation, showcasing key learnings, observations, and contributions made during the internship period.

Every student of M. Sc Quantitative Finance Degree Programme shall undergo an internship in any leading Bank, Financial Institution, Stock Market, Investment Bank, Insurance Companies, Merchant Banking and Stock broking companies for a period of 6 weeks during summer vacation (May & June) under the guidance of a Faculty Member in the Department. Once guides are allotted to the students, the students should contact the respective guides periodically and get necessary guidance and feedback on the project work.

Company should be identified by student as well as by the Department at the end of second semester examinations and it should be communicated to the department, the name of the company in which he/she is undergoing the project, the exact title of the project, the name of the Company Guide and his contact number etc. In the first week of July, all the students have to give a presentation about their observations made by them in internship. Students have to follow a detailed guidelines being circulated by the department in the preparation of internship report. At the end of the internship period, every student shall submit a structured internship report within 15 days from the date of the completion of the project period.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	XX	XXX	XX	XX
CO2	XXX	XXX	XXX	XX	X
CO3	XX	XXX	XXX	X	XX
CO4	X	XX	XX	XXX	XX
CO5	XX	XX	XX	XX X	XXX

MSQF 535: GLOBAL FINANCIAL MANAGEMENT CREDITS: 4

Course Objectives: The aim of this course is to provide an understanding of the fundamental concepts and managerial issues pertaining to international finance.

Course Outcomes (COs):

CO1: Understand the global financial landscape: Comprehend the roles and functions of international financial institutions and development banks like the World Bank, IMF, and ADB, as well as the process of internationalization.

CO2: Analyze foreign exchange markets and rates: Explain the mechanisms of the foreign exchange market, including spot and forward markets, cross-rates, bid-ask spreads, and apply theories determining exchange rates.

CO3: Manage foreign exchange exposure: Evaluate and manage various types of foreign exchange exposure (translation, transaction, and economic) using different methods and strategies.

CO4: Assess multinational firm's financial management: Analyze foreign direct investment decisions, cost of capital, capital structure, budgeting, cash management, country risk, and international taxation for multinational firms.

CO5: Evaluate international financing operations: Understand the characteristics and instruments of Eurocurrency markets, interest rate and currency swaps, depository receipts (GDR and ADR), and the implications of the Euro for India.

Unit-I

International Finance – overview- International Financial Institutions/Development Banks – World Bank – IBRD – IDA – IFC – MIGA – International Monetary Fund – Special Drawing Rights – Asian Development Bank – Internationalization process.

Unit-II

The Foreign Exchange Market – SWIFT – Arbitrage – Spot market – Forward market – Cross rates of exchange – Bid – Ask spreads – Balance of payments – Foreign exchange rates – Theories of Foreign Exchange Rate.

Unit-III

Foreign exchange exposure and management – Management of translation exposure – Methods – Management of transaction exposure – Management of economic exposure - Methods – Strategies.

Unit-IV

Financial Management of the Multinational Firm – Foreign direct investment – Cost of capital and capital structure of the multinational firm – Multinational capital budgeting – Multinational cash management – Country Risk Analysis – International Taxation.

Unit-V

Financing Foreign Operations – Eurocurrency markets – Instruments – Interest rate swaps – Currency swaps and its pricing – Depository receipts – GDR and ADR – Euro and its implications for India.

Books for References:

1. Buckley, A.(1992): Multinational Finance, Prentice Hall, New Jersy.
2. Levich., R.M (2001): International Financial Markets: Prices and Policies, McGraw Hill, New York.
3. Vij, M.(2001): Multinational Financial Management, Excel Books, New Delhi.
4. Shapiro, A.C.(1996): Multinational Financial Management, Prentice hall of India, New Delhi.
5. Apte, P.G.(1999): International Financial Management, Tata McGraw Hill Publishing Company Ltd.
6. Jain, P.K., et.al (1998): International Financial Management, Macmillan, New Delhi.
7. Eun, C.S. and Resnick, B.G.(2001): International Financial Management,Irwin Mcgraw Hill, Singapore.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	X	X	X	XX
CO2	XX	X	XX	X	X
CO3	X	XX	XX	X	X
CO4	XX	X	XX	XX	X
CO5	XXX	X	X	XX	XX

MSQF 536: MACHINE LEARNING TECHNIQUES CREDITS: 4

Course Objectives:

- *To provide knowledge of machine learning techniques*
- *This course familiarizes in supervised and unsupervised learning*

Course Outcomes (COs):

CO1: *Learn the basic concepts of machine learning*
CO2: *Understand the various supervised learning methods*
CO3: *Knowledge of unsupervised learning algorithms*
CO4: *Application of dimensionality reduction methods*
CO5: *Study the fundamental concepts of Artificial Neural Network*

Unit I

Data Pre-processing - Different types of variables – Preparing training and test datasets – Missing Value Imputation - Imbalance in categorical data

Unit II

Supervised Learning – Framework of Regression – Model – Loss Function – Logistic Regression – K Nearest Neighbour (KNN) – Naïve Bayes - Tree based models – Decision Tree – Random Forest – Support Vector Machine

Unit III

Unsupervised Learning – Cluster Analysis - Similarity and Dissimilarity measures – Proximity measures - Types of clustering – Hierarchical clustering: Agglomerative methods and Divisive methods – Non-hierarchical clustering: k-means clustering algorithm – Density based clustering - DBSCAN

Unit IV

Dimensionality Reduction - Principal Component Analysis – Singular Value Decomposition (SVD) – Reinforcement Learning – Recurrent Reinforcement Learning - Bagging, Boosting, XGBOOST

Unit V

Artificial Neural networks – Fundamental concepts on neural networks – Activation Function - Multilayer Feedforward Networks – Convolutional Neural Network – Recurrent Neural Network

Books for reference

1. Hao Ni, Xin Dong, Jinsons Zheng, Guangxi Yu (2021), An Introduction to Machine Learning in Quantitative Finance, World Scientific.
2. Mathew F Dixon, Igor Halperin, Paul Bilokon (2020), Machine Learning in Finance: From Theory to Practice, Springer.
3. Saurav Singla (2020), Machine Learning For Finance, BPB Publications.

COs	PO1	PO2	PO3	PO4	PO5
CO1	X	XXX	XXX	XX	X
CO2	XX	XXX	XXX	XX	X
CO3	XX	XXX	XXX	XX	X
CO4	XX	XX	XXX	XXX	X
CO5	XX	XX	XX	XXX	X

MSQF 542: FINANCIAL ENGINEERING AND DERIVATIVES CREDITS: 3

Course Objectives:

- *A broad range of derivative products are examined with a primary focus on how to use these for the management of financial risks.*
- *The course introduces standard models of pricing forward, futures and options on diverse underlying assets*.
- *The course then explores hedging methods to conduct risk management for business operations, speculative trades, and issued financial instruments.*
- *After completing this course students will be familiar with derivatives valuation and their use in risk management.*

Course Outcomes (COs):

C01. Understand the concept, scope, and significance of financial engineering and its role in modern financial systems.

C02. Explain the structure, functions, and types of financial derivatives and assess their relevance in global markets.

C03. Analyze the mechanics and application of futures contracts and develop hedging strategies using futures.

C04. Apply option pricing models and construct various option trading strategies to manage risk and returns.

C05. Evaluate different types of swap agreements and understand their valuation, mechanics, and associated risks.

Unit I: Introduction

Introduction to Financial Engineering-Meaning, scope and Need-Tools of Financial Engineering-Financial Engineering and Financial Analysis-Factors Contributing to the Growth of Financial Engineering-Financial Engineering process

Unit II: Financial Derivatives

Introduction to Derivatives - Meaning- Definition- function and types of Derivatives - Derivatives Market in India and other countries - OTC and New Financial Derivatives emerging in international financial markets.

Unit III: Futures

Forward and Futures Contracts - Futures Markets- Mechanics of Futures Markets - Long and Short of Financial Futures-Closing out futures positions- Specification of Futures Contracts- Hedging strategies using futures - Convergence of futures price to spot price -MTM - Clearing House Arrangement - Stock index futures.

Unit IV: Options

Options Contract - Meaning - Types of Options - Option pricing models - Binomial model - Black-Scholes model - Differences between Futures and Options Contract - Options Trading strategies - Covered call - Protective put - Spreads - Bull spreads - Bear spreads - Butterfly spreads - Calendar spreads - Straddle – Strips and straps - Strangles - Put-call parity theorem.

Unit V: Swaps

Swaps-Meaning - Types - Interest Rate Swap - Currency Swaps - Valuation - mechanics of operation - Credit Risk and Swaps -

Books for Reference

1. Mehrotra, S. (2025). *Financial derivatives and risk management: Derivative strategies*. Pen and Paper Academy. ISBN: 9788198456595.
2. Radhika, R., Balakumar, V., Elayamurugan, S., & Krishnan, C. G. (2025). *Financial risk management and derivatives*. Paradox International Publications.
3. Kumar, S. (2021). *Financial engineering and quantitative risk analytics*. SYBGEN Learning. ISBN: 978-8195131952.
4. Hull, J. (2021). *Options, futures, and other derivatives* (10th ed.). Pearson Education Limited. ISBN: 978-1292410654.
5. Malarvizhi, S. (2024). *Financial derivatives*. Imaginex Inks Publication.

COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	X	X	X	XX
CO2	XXX	X	XX	X	XX
CO3	XX	XX	XXX	XX	X
CO4	XX	XXX	XXX	X	X
CO5	XX	XXX	XX	XX	X